MedBhasha – Exploratory Data Analysis (EDA) Summary

[By team: Train test panic]

1. Dataset Overview

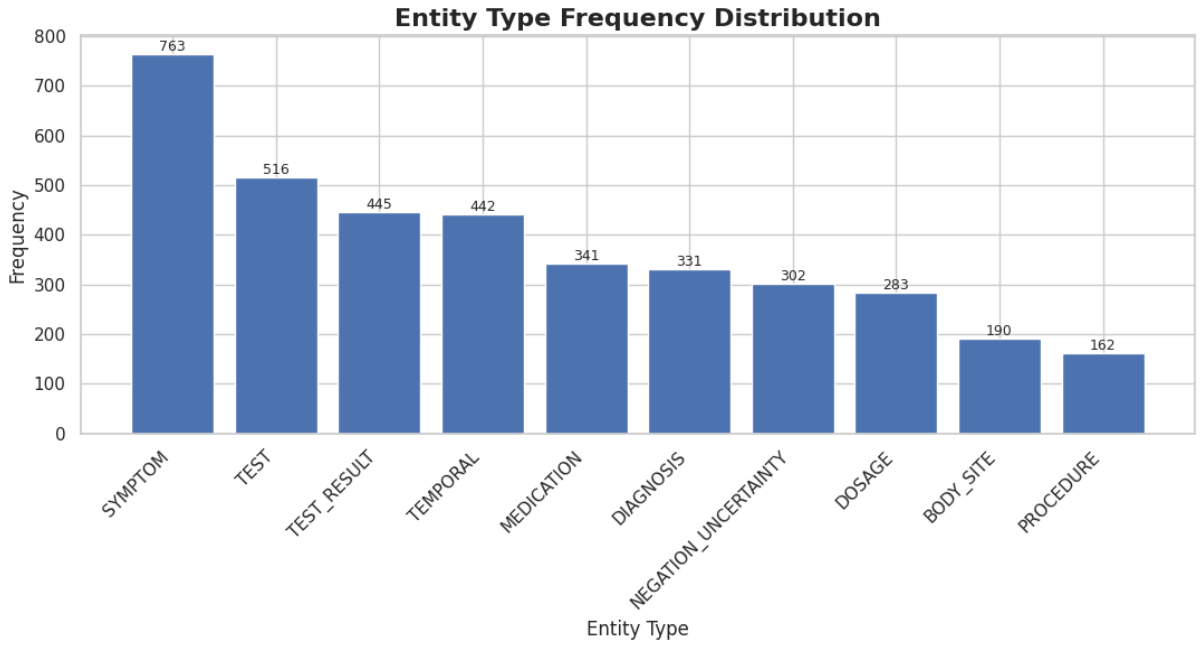
The MedBhasha corpus consists of 300 synthetic Bengali medical notes generated using Claude AI (Anthropic), Sonnet 5.5. The dataset contains 14,452 tokens annotated using the BIO (Beginning, Inside, Outside) tagging scheme.

The corpus includes 10 medical Named Entity Recognition (NER) categories, with a total of 3,775 annotated entity spans.

Dataset Attribute	Value
Total medical notes	300
Total tokens	14,452
Total entity spans	3,775
Entity categories	10
Average tokens per note	48.17
Annotation format	BIO

2. Entity Frequency Distribution

The entity frequency distribution shows the number of occurrences of each medical entity category in the corpus.



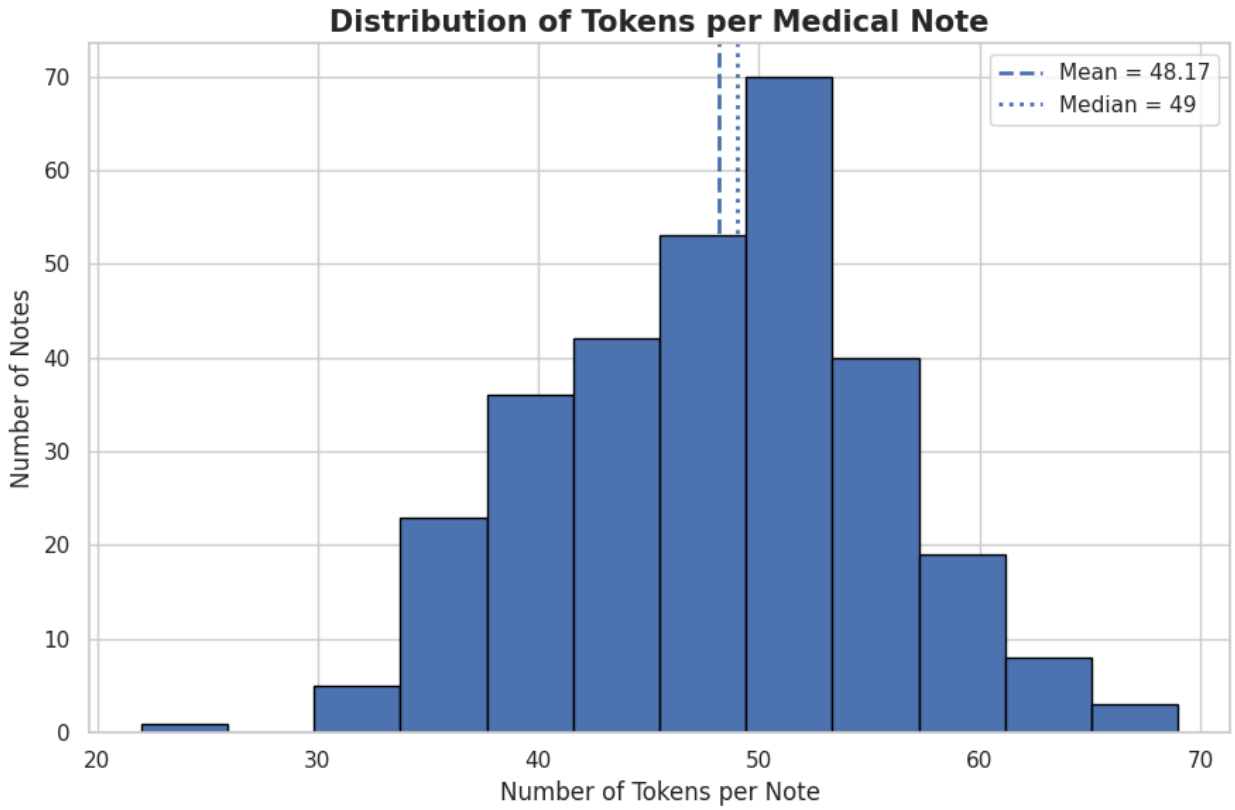
Entity Category	Frequency	Percentage
SYMPTOM	763	20.21%
TEST	516	13.67%
TEST_RESULT	445	11.79%
TEMPORAL	442	11.71%
MEDICATION	341	9.03%
DIAGNOSIS	331	8.77%
NEGATION_UNCERTAINTY	302	8.00%
DOSAGE	283	7.50%
BODY_SITE	190	5.03%
PROCEDURE	162	4.29%
Total	**3,775**	**100%**

Observation: SYMPTOM is the most frequent entity category, accounting for 20.21% of all annotated entities. PROCEDURE is the least frequent, accounting for 4.29%. This indicates that symptom-related information has the highest representation in the corpus.

3. Note-Length Distribution

Note length was calculated by counting the number of tokens in each medical note.

Figure 2: Distribution of Tokens per Note



Statistical Measure	Token Count
Mean	48.17
Median	49
Minimum	22
Maximum	69
Standard Deviation	7.58

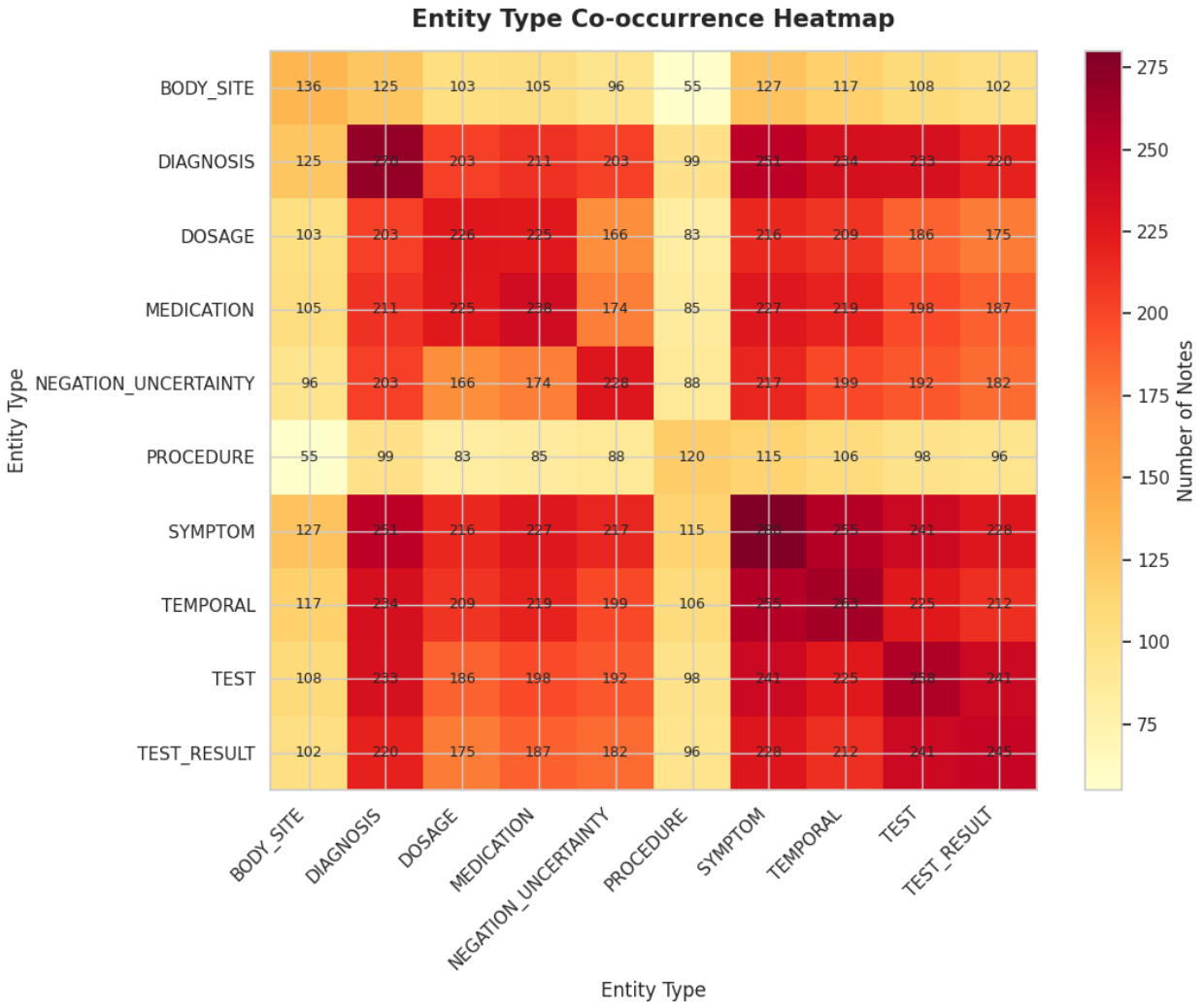
Observation: The notes contain an average of approximately 48 tokens, with lengths ranging from 22 to 69 tokens. The mean and median are close, suggesting that note lengths are generally distributed around a similar central range.

4. Entity Relationship and Co-occurrence Analysis

A basic entity co-occurrence analysis was performed to examine how frequently different entity categories appear together within the same medical note.

A 10 × 10 co-occurrence matrix was constructed. Each cell represents the number of notes containing both corresponding entity categories at least once. The diagonal represents the number of notes containing each individual entity category.

Figure 3: Entity Type Co-occurrence Heatmap



The most frequent entity-type co-occurrences were:

Entity Pair	Number of Notes
SYMPTOM – TEMPORAL	255
DIAGNOSIS – SYMPTOM	251

TEST – TEST_RESULT	241
SYMPTOM – TEST	241
DIAGNOSIS – TEMPORAL	234
MEDICATION – SYMPTOM	227
DOSAGE – MEDICATION	225

Observation: SYMPTOM and TEMPORAL show the highest co-occurrence, appearing together in 255 notes. TEST and TEST_RESULT also frequently occur together, appearing in 241 notes. Similarly, MEDICATION and DOSAGE co-occur in 225 notes. These patterns show how different medical information categories are represented together within the notes.

This analysis measures co-occurrence within the same note and does not establish a direct semantic or clinical relationship between the entities.

5. Key Findings

- The corpus contains 300 medical notes, 14,452 tokens, and 3,775 annotated entity spans.
- SYMPTOM is the most frequent entity category, while PROCEDURE is the least frequent.
- Each note contains an average of 48.17 tokens.
- SYMPTOM and TEMPORAL have the highest observed co-occurrence, appearing together in 255 notes.
- TEST and TEST_RESULT frequently appear together, while MEDICATION and DOSAGE also show notable co-occurrence.

6. Conclusion

The exploratory data analysis provides an overview of the entity distribution, note lengths, and co-occurrence patterns in the MedBhasha corpus. The results highlight the representation of different medical entity categories and their presence together within individual notes.

These findings provide a basic understanding of the corpus structure and can help identify unusual distributions or potential annotation inconsistencies for further inspection before final submission.